Horizontal and vertical dilations

1 Consider the functions $f(x) = \cos x$ and $g(x) = \cos 4x$.

a State the period of $f(x)$ in radians.

b Complete the table of values for $g(x)$.

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$	$\frac{5\pi}{8}$	$\frac{3\pi}{4}$	$\frac{7\pi}{8}$	π
$g(x)$									

c State the period of $g(x)$ in radians.

d Describe the transformation required to obtain the graph of $g(x)$ from $f(x)$.

e Sketch the graph of $g(x)$ for $0 \leq x \leq \pi$.

2 Consider the function $y = 2\cos 3x$.

a State the amplitude of the function.

b Find the period of the function in radians.

c Sketch a graph of the function for $-\pi \leq x \leq \pi$.

3 For each of the following functions:

i State the amplitude.

ii Find the period in radians.

iii Sketch a graph of the function for $0 \leq x \leq 2\pi$.

a $y = \cos 3x$ **b** $y = \sin\left(\frac{x}{2}\right)$ **c** $y = \sin\left(\frac{3x}{4}\right)$ **d** $y = \cos\left(\frac{x}{3}\right)$

e $y = -\cos 4x$ **f** $y = 3\sin(0.25x)$ **g** $y = 2\sin 3x$ **h** $y = -5\sin 3x$

4 Consider the function $y = \sin\left(\frac{2x}{3}\right)$.

a State the amplitude of the function.

b Find the period of the function in degrees.

c Sketch a graph of the function for $0^\circ \leq x \leq 720^\circ$.

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5 A table of values for the the first period of the graph $y = \sin x$ for $x \geq 0$ is given in the first table on the right:

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\sin x$	0	1	0	-1	0

a Complete the second table given with equivalent values for x in the the first period of the graph $y = \sin\left(\frac{x}{4}\right)$ for $x \geq 0$.

x					
$\sin\left(\frac{x}{4}\right)$	0	1	0	-1	0

b Hence, state the period of $y = \sin\left(\frac{x}{4}\right)$.

6 Consider functions of the form $y = \tan bx$.

a Complete the given table identifying the periods of the functions.

Function	Period
$\tan x$	π
$\tan 2x$	
$\tan 3x$	
$\tan 4x$	

b State the period of $y = \tan bx$.

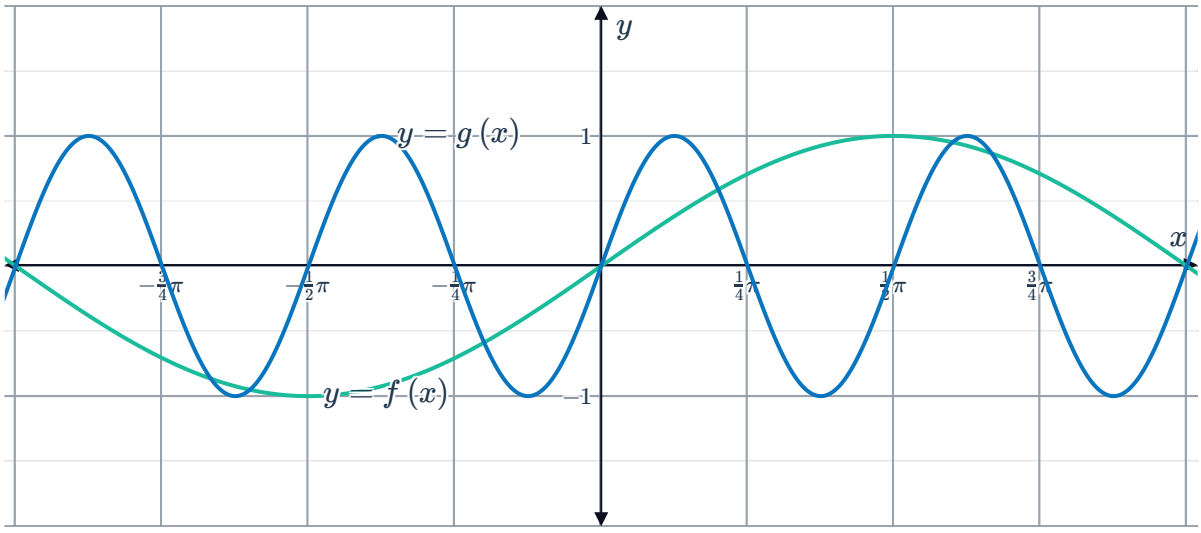
c As the value of b in $\tan bx$ increases, does the period become shorter or longer?

7 State the period of the following functions:

a $y = \cos 3x$ **b** $y = \sin(2\pi x)$ **c** $y = \tan 2x$ **d** $y = \tan\left(\frac{x}{4}\right)$

e $y = \cos(\pi x)$ **f** $y = \tan 0.75x$ **g** $y = \tan\left(\frac{2x}{3}\right)$ **h** $y = \tan\left(\frac{\pi x}{2}\right)$

8 The functions $f(x)$ and $g(x) = f(kx)$ have been graphed on the same set of axes below.



a Describe the transformation required to obtain the graph of $g(x)$ from the graph of $f(x)$.

b Find the value of k.

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9 For each of the following functions:

i Find the y-intercept.

ii Find the period of the function in radians.

iii Find the distance between the asymptotes of the function.

iv State the first asymptote of the function for $x \geq 0$

v State the first asymptote of the function for $x \leq 0$

vi Sketch a graph of the function for $-\pi \leq x \leq \pi$.

a $y = \tan 2x$ **b** $y = \tan 3x$ **c** $y = \tan\left(\frac{x}{2}\right)$ **d** $y = \tan(-4x)$

10 Consider the equation $y = \tan 9x$.

a State the period of the equation in radians.

b Sketch the graph of the equation $y = \tan 9x$ for $0 \leq x \leq \pi$

11 Consider the function $y = \tan 7x$.

a Complete the given table of values for the function.

b Graph the function for $-\frac{5\pi}{28} \leq x \leq \frac{5\pi}{28}$.

x	$-\frac{\pi}{28}$	0	$\frac{\pi}{28}$	$\frac{3\pi}{28}$	$\frac{\pi}{7}$	$\frac{5\pi}{28}$
y						

12 A function in the form $f(x) = \tan bx$ has adjacent x-intercepts at $x = \frac{13\pi}{6}$ and $x = \frac{9\pi}{4}$.

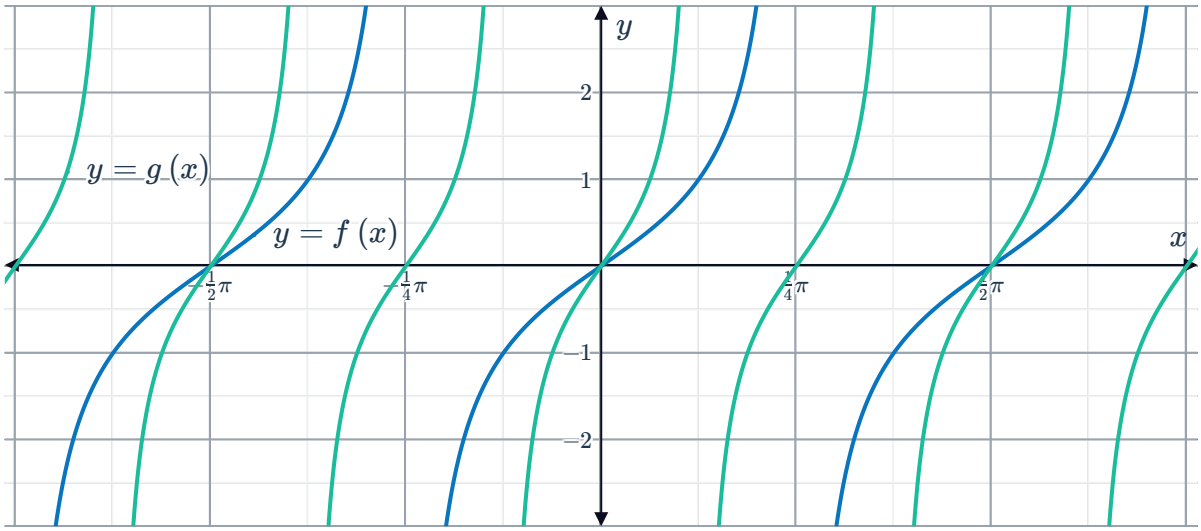
a State the equation of the asymptote lying between the two x-intercepts.

b Find the period of the function.

c State the equation of the function.

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13 Consider the graph of $f(x) = \tan(\alpha x)$, and the graph of $g(x) = \tan(\beta x)$ below:



Which is greater: α or β? Explain your answer.

14 A table of values for the the first period of the graph $y = \tan x$ for $x \geq 0$ is given in the first table on the right:

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$\tan x$	0	1	undefined	-1	0

a Complete the second table given with equivalent values for x in the the first period of the graph $y = \tan\left(\frac{x}{6}\right)$ for $x \geq 0$.

x					
$\tan\left(\frac{x}{6}\right)$	0	1	undefined	-1	0

b Hence, state the period of $y = \tan\left(\frac{x}{6}\right)$.

d State the first three asymptotes of the function for $x \geq 0$.

e State the first asymptote of the function for $x \leq 0$.

Phase shifts

15 Consider the function $y = \sin\left(x + \frac{\pi}{2}\right)$.

a Find the maximum value of the function.

b Find the minimum value of the function.

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16 Consider the function $f(x) = \cos x$ and $g(x) = \cos\left(x - \frac{\pi}{2}\right)$.

a Complete the table of values for both functions:

b Describe the transformation required to obtain the graph of $g(x)$ from $f(x)$.

c Sketch a graph of $g(x)$ for $0 \leq x \leq 2\pi$.

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(x)$					
$g(x)$					

17 For each of following functions:

i State the amplitude of the function.

ii Describe the transformation required to obtain the graph of $y = f(x)$ from the graph of $y = \sin x$.

iii Sketch a graph of the function for $-\pi \leq x \leq \pi$.

a $f(x) = \sin\left(x - \frac{\pi}{2}\right)$

b $f(x) = \sin\left(x + \frac{\pi}{4}\right)$

18 For each of following functions:

i State the amplitude of the function.

ii Describe the transformation required to obtain the graph of $y = f(x)$ from the graph of $y = \cos x$.

iii Sketch a graph of the function for $-\pi \leq x \leq \pi$.

a $f(x) = \cos(x + \pi)$

b $f(x) = \cos(x - \pi)$

c $f(x) = \cos\left(x + \frac{3\pi}{2}\right)$

19 Find the values of c in the region $-2\pi \leq c \leq 2\pi$ that make the graph of $y = \sin(x - c)$ the same as the graph of $y = \cos x$.

20 Sketch the graph of the following for $-2\pi \leq x \leq 2\pi$.

a $y = \sin\left(x + \frac{\pi}{2}\right)$

b $y = \cos\left(x + \frac{\pi}{2}\right)$

21 True of false: The graph of $f(x) = \tan\left(x - \frac{\pi}{6}\right)$ is a phase shift of $y = \tan x$ by $\frac{\pi}{6}$ units to the right.

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22 For each of the functions below, state the horizontal translation required to obtain the graph of $g(x)$ from the graph of $f(x) = \tan x$:

a $g(x) = \tan\left(x + \frac{\pi}{2}\right)$

b $g(x) = \tan\left(x - \frac{\pi}{5}\right)$

c $g(x) = \tan(x - 1.5)$

d $g(x) = \tan\left(x + \frac{3\pi}{4}\right)$

23 The function $f(x) = \tan\left(x - \frac{\pi}{6}\right)$ is to be graphed on the interval $\left[\frac{2\pi}{3}, \frac{8\pi}{3}\right]$.

a Find the asymptotes of the function that occur on this interval.

b Find the x-intercepts of the function that occur on this interval.

c Sketch a graph of the function $f(x) = \tan\left(x - \frac{\pi}{6}\right)$ over the interval $\left[\frac{2\pi}{3}, \frac{8\pi}{3}\right]$.

24 For each of the following functions:

i Find the y-intercept.

ii Find the period of the function.

iii Find the distance between the asymptotes of the function.

iv State the first asymptote of the function for $x > 0$.

v State the first asymptote of the function for $x \leq 0$.

vi Sketch a graph the function for $-\pi \leq x \leq \pi$.

a $y = \tan\left(x - \frac{\pi}{2}\right)$

b $y = \tan\left(x - \frac{\pi}{6}\right)$

c $y = \tan\left(x + \frac{\pi}{3}\right)$

25 Consider the function $f(x) = \tan\left(x - \frac{\pi}{7}\right)$.

a Compared to the equation $y = \tan x$, state the phase shift of $f(x)$.

b State the equations of the first four asymptotes of $f(x)$ to the right of the origin.

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Combined translations and dilations

26 Consider the function $y = \cos 3x + 2$.

- **a** Find the period of the function, giving your answer in radians.
- **b** State the amplitude of the function.
- **c** Find the maximum value of the function.
- **d** Find the minimum value of the function.
- **e** Sketch a graph of the function for $0 \leq x \leq 2\pi$.

27 For each of the following functions:

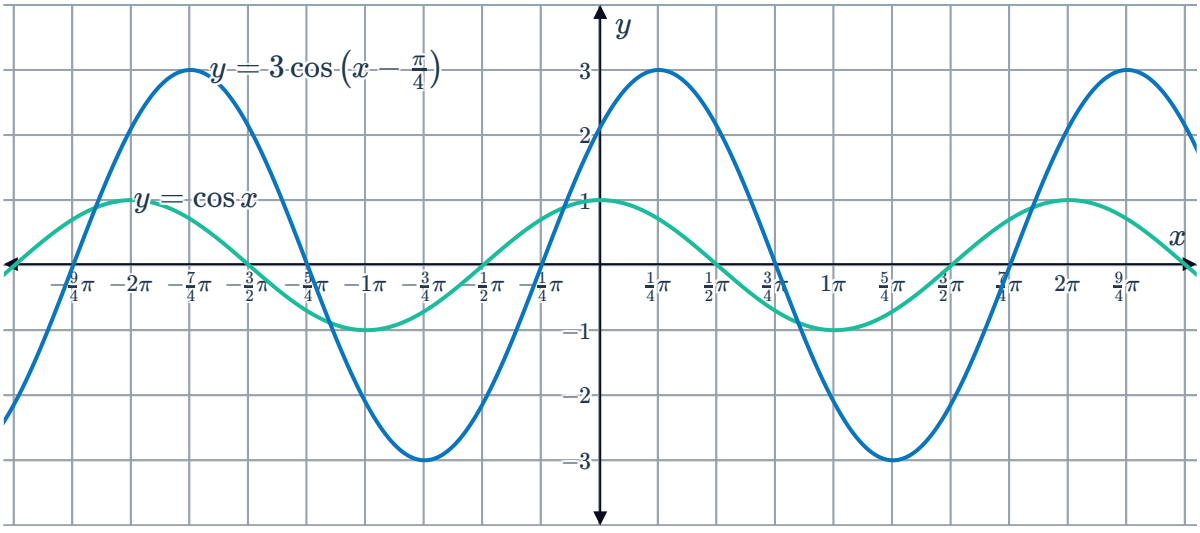
- **i** State the domain of the function.
- **ii** State the range of the function.
- **iii** Sketch a graph of the function for $-\pi \leq x \leq \pi$.

- **a** $y = \sin 2x - 2$
- **b** $y = -5 \sin 2x$
- **c** $y = \sin\left(\frac{x}{3}\right) + 5$
- **d** $y = \cos\left(\frac{x}{2}\right) - 3$

28 Sketch a graph of each of the following the functions for $0 \leq x \leq 2\pi$:

- **a** $y = 4\cos\left(x - \frac{\pi}{2}\right)$
- **b** $y = \tan\left(\frac{\pi}{2} - x\right)$
- **c** $y = \tan\left(2\left(x - \frac{2\pi}{3}\right)\right)$

29 Consider the graphs of $y = \cos x$ and $y = 3\cos\left(x - \frac{\pi}{4}\right)$.



Describe the series of transformations required to obtain the graph of $y = 3\cos\left(x - \frac{\pi}{4}\right)$ from the graph of $y = \cos x$.

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30 Consider the function $f(x) = -5\sin\left(x + \frac{\pi}{3}\right)$.

a Describe the series of transformations required to obtain the graph of $f(x)$ from the graph of $y = \sin x$.

b Sketch a graph of $y = f(x)$ for $-2\pi \leq x \leq 2\pi$.

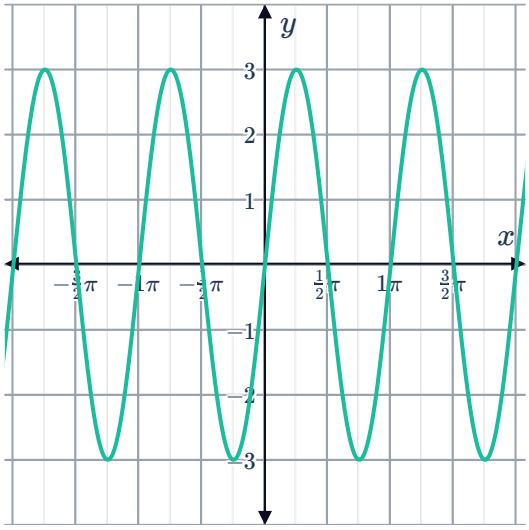
31 The graph of $y = \cos x$ undergoes the series of transformations below:

- The graph is reflected about the x-axis.
- The graph is horizontally translated to the left by $\frac{\pi}{6}$ radians.
- The graph is vertically translated upwards by 5 units.

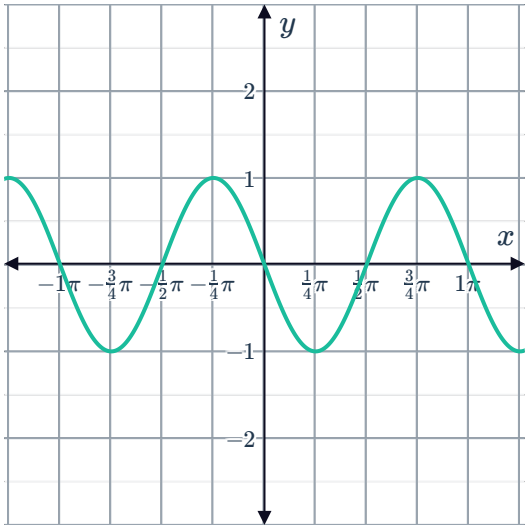
Find the equation of the transformed graph in the form $y = a\cos(x + c) + d$, where c is the lowest positive value in radians.

32 For each of the following graphs, find the equation of the function given that it is of the form $y = a\sin bx$ or $y = a\cos bx$, where b is positive:

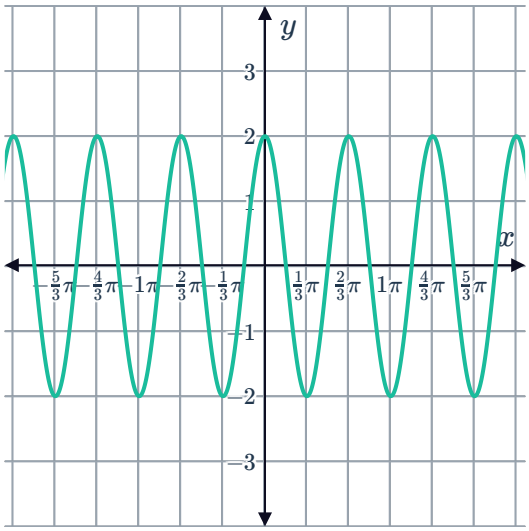
a



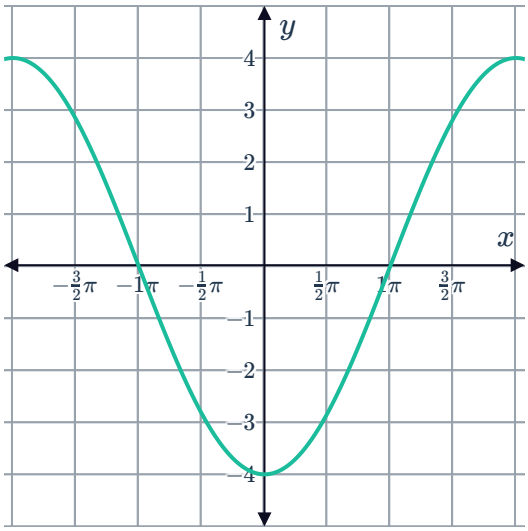
b



c

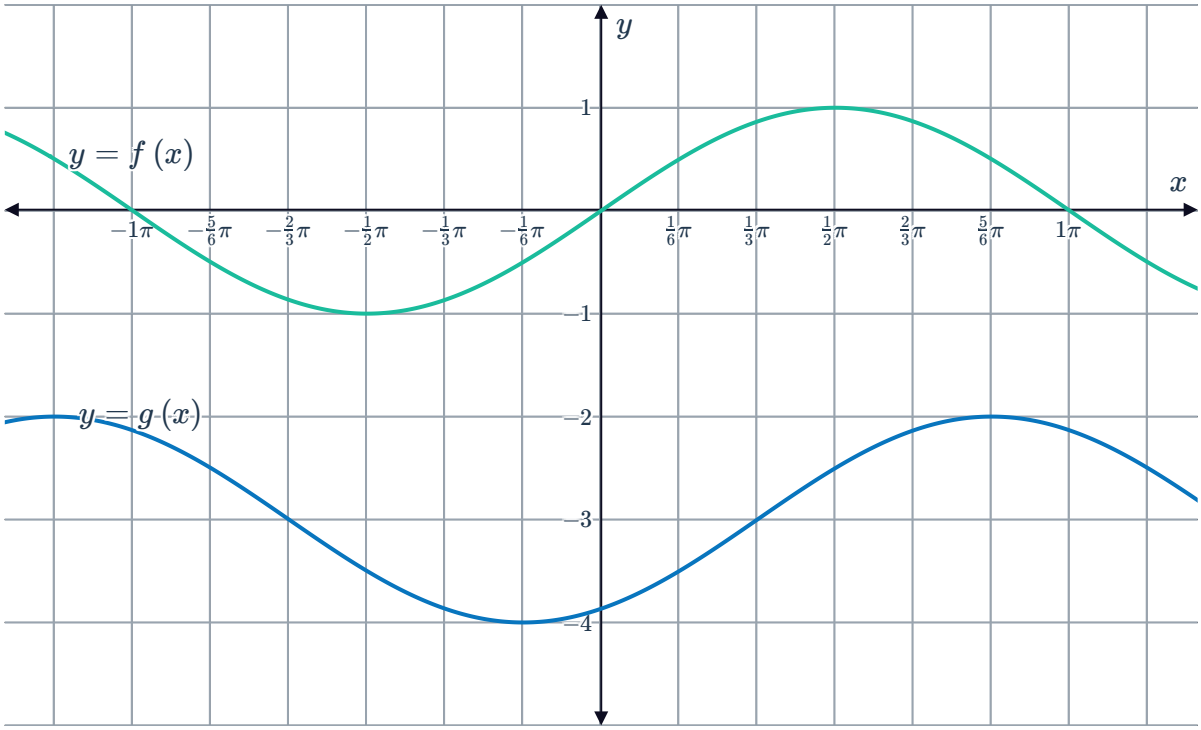


d



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33 The functions $f(x)$ and $g(x) = f(x-k) - j$ have been graphed on the same set of axes below:



a Find the value of j.

b Find the smallest positive value of k.

c Describe the series of transformations required to obtain the graph of $g(x)$ from the graph of $f(x)$.

34 Describe the series of transformations required to obtain the graph of $y = \sin\left(x - \frac{\pi}{4}\right) + 2$ from the graph of $y = \sin(x)$.

35 State whether the following functions represent a change in the period from the function $y = \sin x$:

a $y = \sin(5x)$ b $y = \sin(x-5)$ c $y = 5\sin x$ d $y = \sin\left(\frac{x}{5}\right)$

e $y = \sin x + 5$

36 For each of the following functions:

i State the amplitude of the function.

ii Describe the transformation required to obtain the graph of $y = f(x)$ from the graph of $y = \sin(x)$.

iii Sketch a graph of the function for $0 \leq x \leq 2\pi$.

a $f(x) = 4\sin(x-\pi)$ b $f(x) = -4\sin\left(x + \frac{\pi}{2}\right)$

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37 To obtain the graph of $y = \tan\left(2\left(x - \frac{\pi}{4}\right)\right)$ from $y = \tan x$, two transformations were applied:

- Horizontal translation of $\frac{\pi}{4}$ units to the right.
- Horizontal dilation by a scale factor of $\frac{1}{2}$.

a State the transformation that was applied first.

b Sketch the graph of each of the following functions for $-\pi \leq x \leq \pi$:

i $y = \tan 2x$

ii $y = \tan\left(2\left(x - \frac{\pi}{4}\right)\right)$

38 Consider the function $f(x) = \cos\left(2x + \frac{\pi}{3}\right)$.

a Describe the series of transformations required to obtain the graph of $f(x)$ from the graph of $y = \cos x$ given that a horizontal dilation was performed first.

b Sketch the graph of $y = f(x)$ for $0 \leq x \leq 2\pi$.

39 Describe the series of transformations required to obtain the graph of $y = \tan\left(2x - \frac{\pi}{4}\right)$ from $y = \tan x$ given that a horizontal dilation was performed first.

40 Consider the function $y = \cos(3x + \pi)$.

a Find the period of the function.

b Describe the series of transformations required to obtain the graph of $y = f(x)$ from the graph of $y = \cos x$ given that a horizontal dilation was performed first.

c Sketch a graph of the function for $-\pi \leq x \leq \pi$.

41 For each of the following functions:

i Find the period.

ii State the amplitude.

iii Find the maximum value.

iv Find the minimum value.

v Graph the function for $0 \leq x \leq 2\pi$.

a $y = \cos 3x + 2$

b $y = 2\cos\left(x - \frac{\pi}{2}\right) + 3$

c $y = 3\sin\left(x - \frac{\pi}{3}\right) + 2$

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42 Consider the function $y = -4\tan\frac{1}{5}\left(x + \frac{\pi}{4}\right)$.

a Find the period of the function.

b Describe a series of transformations to obtain the graph of $y = f(x)$ from the graph of $y = \tan x$.

c Find the range of the function.

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